Future Flow

WATER QUALITY TESTER

People behind this project:

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SQUAD

ABOUT THE PROJECT

This project is designed to monitor and analyze the quality of water

IT MEASURES IMPORTANT WATER PARAMETERS SUCH AS:

PH LEVEL

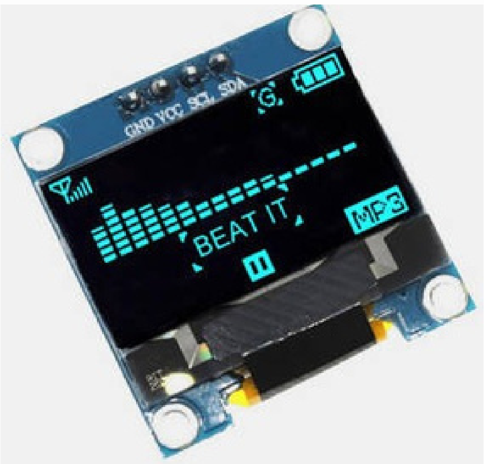
TURBIDITY

TEMPERATURE

TOTAL DISSOLVED SOLIDS (TDS)

LIST OF COMPONENTS:

1.ESP32 Dev Board(ESP32-WROOM-32)
2.OLED Display0.96"(I2C, SSD1306)
3.pH Sensor Module+Probe (with BNC connector)
4.Turbidity Sensor Module (Analog Output)
5.DS18B20 Wa terproof Temperature Sensor
6.4.7kΩ Resistor
7.Active Buzzer (3.3V–5V)
8.Breadboard (Medium size)
9.Jumper Wires (Male-Male + Male-Female)



Datasheet of Components:

1.ESP32 Dev Board (ESP32-WROOM-32):

- Core Specifications:

Processor : Xtensa® Dual-Core 32-bitLX6
Clock Frequency : Up to 240 MHz
SRAM : 520 KB
Flash Memory : 4 MB (Typical for WROOM modules)
Wireless : Wi-Fi (802.11 b/g/n) & Bluetooth (v4.2 BR/EDR and BLE)
Operating Voltage : 2.2V to 3.6V (Module) / 5V (via Micro-USB)

- Pinout& Peripherals:

ADC (Analog to Digital): 12-bit resolution across 18 channels. Great for sensors.
DAC (Digital to Analog): 2 channels (8-bit) for generating true analog voltages.
Capacitive Touch: 10 GPIOs have built-in touch sensors (no buttons needed!).
Communication: 3 x UART, 3 x SPI, 2 x I2C, and CAN bus 2.0.
PWM: 16 independent channels (perfect for controlling many servos or LEDs).

Datasheet of Components:

1.ESP32 Dev Board (ESP32-WROOM-32):

- Absolute Maximum Ratings:

VDD (Supply Voltage): -0.3V to +3.6V

IO (Current per GPIO): 40 mA (recommended is 12-20 mA)

Total Current (Internal Regulator): ~500 mA to 1.2A (depending on Wi-Fi usage spikes)

- Power Consumption (Modes):

Active Mode: 160–240 mA (Wi-Fi/BT on)

Light Sleep: 0.8 mA

Deep Sleep: ~10 μA (The RTC is still on; perfect for battery-powered weather stations)

- Typical Applications:

Web Servers: Hosting a small HTML page to control lights in your room

Datasheet of Components:

2.OLED Display 0.96" (I2C, SSD1306):

- Physical& DisplaySpecifications:

Display TechnologyOLED (Organic Light Emitting Diode)
Resolution128 x 64 pixels
Diagonal Size0.96 inch
Viewing Angle> 160° (stays clear even from the side)
Colors
Monochrome (usually Blue, White, or Yellow/Blue bi-color)

- Electrical Characteristics:

Operating Voltage (VCC): 3.3V to 5V (Most modules have an onboard regulator)
Logic Level: Compatible with both 3.3V (ESP32/STM32) and 5V (Arduino Uno)
Power Consumption:

1.Normal operation: ~20 mA (Highly dependent on how many pixels are lit).
2.All pixels OFF: ~10 μA.
3.All pixels ON: ~25 mA to 30 mA.

Datasheet of Components:

2.OLED Display 0.96" (I2C, SSD1306):

- I2CPinout (4-Pin Version):

1 .GND:Ground
2.VCC:Power (3.3V or 5V)
3.SCL:Serial Clock Line (I2C Clock)
4.SDA:Serial Data Line (I2C Data)

- Internal Memory (GDDRAM):

Each bit corresponds to one pixel (1=ON,0=OFF).
The memory is divided into 8 Pages (Page 0 to Page 7).
Each page contains 8 rows and 128 columns.

- Software Integration:

Arduino IDE: Adafruit_SSD1306 and Adafruit_GFX.
MicroPython: ssd1306.py library.

Datasheet of Components:

3.pH Sensor Module + Probe (with BNC connector):

- Module Specifications (TheSignalConditioner):

Input Voltage (VCC) : 5.0V ± 0.2V Working Current : 5–10 mA Output Type : Analog Voltage (Signal Pin Po) Output Range : 0V to 5V (Ideally 2.5V at pH 7.0) Adjustments : 2 Blue Potentiometers (Offset & Threshold) Module Size : 42mm x 32mm pH Probe Specifications: Measurement Range : 0 – 14 pH Accuracy : ± 0.1 pH (at 25°C) Response Time : ≤ 1 minute Connector Type : BNC (Standard) Cable Length : ~75 cm to 1 meter Probe Life : ~6 months to 1 year (depending on usage)

Datasheet of Components:

3.pH Sensor Module + Probe (with BNC connector):

- PinConfiguration &HardwareControls:

V+ / VCC: Connect to 5V.

G / GND: Connect to Ground.

Po / Analog Out: Connect to an Analog Pin (e.g., GPIO 34 on ESP32).

Do / Digital Out: High/Low output based on a threshold set by the second potentiometer.

To : Temperature output (Note: often not connected or functional on generic boards).

Datasheet of Components:

4.Turbidity Sensor Module (Analog Output)

- Electrical Specifications:

Operating Voltage (VCC) : 5.0V DC
Operating Current : 11mA to 40mA (MAX)
Response Time : < 500 ms
Detection Range : 0 to 1000 NTU (typical) / 0 to 4550 NTU (max)
Output Type : Analog (0-4.5V) & Digital (High/Low)
Operating Temperature : 5°C to 90°C

- Pinout Configuration:

1(Pin) : VCC : 5V Powe r Supply
2(Pin) : GND : Ground
3(Pin) : Aout / A0 : Analog Output (Voltage decreases as water gets dirtier)
4(Pin) : Dout / D0 : Digital Output (Threshold set by potentiometer)

- Working Principle & Mapping:

Clear Water: High transmittance → High Voltage (~4.1V to 4.5V).
Very Turbid Water: Low transmittance → Low Voltage (~0V to 2.5V).

Datasheet of Components:

5.DS18B20 Waterproof Temperature Sensor

- CoreTechnical Specifications:

Operating Voltage : 3.0V to 5.5V
Temperature Range : -55°C to +125°C (-67°F to +257°F)
Accuracy : ± 0.5°C (from -10°C to +85°C)
Resolution : User-selectable from 9 to 12 bits
Conversion Time : 750ms (at 12-bit resolution)
Unique ID : Each sensor has a unique 64-bit serial code

- Pinout & Wiring (Waterproof Probe):

Red(Wire Colour) : VC C : Power Supply (3.3V o r 5V)
Black(Wire Colour) : GND : Ground
Yellow (or Blue)(Wire Colour) : DATA : Digital 1-Wire Signal

Datasheet of Components:

6.Resistor 4.7kΩ:

- Specifications (Standard CarbonFilm):

Resistance : 4.7kΩ (4700Ω)
Power Rating : 1/4W (0.25W) is standard; 1/8W or 1/2W also common
Tolerance : ±5% (Gold band) or ±1% (Brown band)
Max Working Voltage : 250V to 350V
Temperature Coefficient : -500 to +350 ppm/°C

- Color Code Identification:

4-Band Resistor (±5% Tolerance):
Band 1 (1st Digit): Yellow (4)
Band 2 (2nd Digit): Violet (7)
Band 3 (Multiplier): Red (×100)
Band 4 (Tolerance): Gold (±5%)
5-Band Resistor (±1% Precision):
Band 1: Yellow (4)
Band 2: Violet (7)
Band 3: Black (0)
Band 4: Brown (×10)
Band 5: Brown (±1%)

Datasheet of Components:

6.Resistor 4.7kΩ:

- Physical Dimensions:

Body Length: ~6.3 mm
Body Diameter: ~2.4 mm
Lead Diameter: ~0.6 mm

Datasheet of Components:

7.Active Buzzer (3.3V–5V)

- Technical Specifications:

Operating Voltage : 3.3V to 5.0V DC(Works perfectly with ESP32)
Current Consumption : ~20mA to 30mA (While active)
Resonant Frequency : 2300 Hz ± 300 Hz (A high-pitched beep)
Sound Output (SPL) : ≥ 85dB at 10cmDriving MethodDC Voltage (Digital High/Low)Working
Temp : -20°C to +70°C

- Pinout (Module Version):

1(Pin) : VCC : 3.3V or 5V Power
2(Pin) : GND : Ground
3(Pin) : I/O (S) : Signal Pin (High = Sound, Low = Quiet)

Datasheet of Components:

8.Breadboard (Medium size):

- TechnicalSpecifications (Standard 400-Point):

Total Tie Points : 400 (300 in the main area + 100onpower rails) Contact Material : Phosphor Bronze (Nickel-plated) Body Material : ABS Plastic (Heat distortion at ~84°C) Hole Pitch : 2.54 mm (Standard 0.1") — Matches IC pins Dimensions : ~82 mm x 55 mm Insertion Life : ~50,000 cycles

- Electrical Ratings:

Voltage Rating : Typic ally rated up to 300V (but practically used for <4 8V).
Current Rating : 1A to 2A max per strip.
Contact Resistance : ~0.1 Ω.
Parasitic Capacitance : 2 pF to 25 pF between adjacent rows.

Datasheet of Components:

9.Jumper Wires (Male-Male + Male-Female)

- Technical Specifications:

Wire Gauge : 24 AWG to 26 AWG (Standard)
Core Material : Tinned Copper or Copper-Clad Aluminum (CCA)I
nsulation : PVC (Polyvinyl Chloride)
Connector Type : 2.54mm (0.1") Pitch DuPont Connector
Current Rating : ~1.5A to 2A (Max)
Voltage Rating : Up to 300V (Practically used for 3.3V–12V)
Lengths : 10cm, 20cm (most common), or 30cm

PRICING TABLE:

ESP32 Dev Board (ESP32-WROOM-32)
₹400-₹600

OLED Display 0.96" (I2C, SSD1306)
₹200-₹300

PRICING TABLE:

pH Sensor Module + Probe (with BNC connector)
₹1,000-₹1500

Turbidity Sensor Module
(Analog Output)
₹400-₹600

PRICING TABLE:

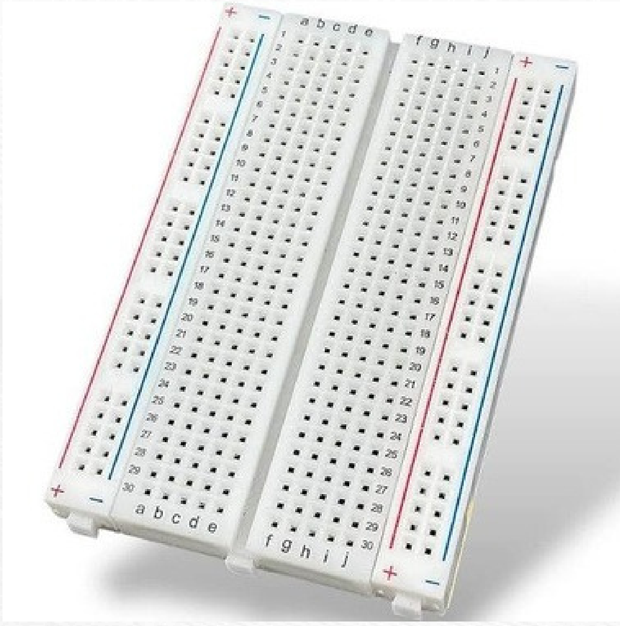
DS18B20 Waterproof Temperature Sensor
₹150-₹200

4.7kΩ Resistor
₹10-₹20

PRICING TABLE:

Active Buzzer (3.3V–5V)
₹30-₹100

Breadboard (Medium size)
₹70-₹140



PRICING TABLE:

Jumper Wires (Male-Male + Male-Female)
₹60-₹120

PRICING TABLE:

Total Estimated Price:
₹2100-₹3800

Referances:

1.Gemini
2. ChatGPT
3. Chrome
4. Alldatasheet.com
5.Solomon systech
6. Componentstree.com

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Thank you